```
decision_id=7b3f… result=DENY
reason=principal not in Project::"alpha".members for Action::"doc:ed
it"
 on Document::"alpha/roadmap.md"
```

With a small internal dashboard, engineers filtered by `decision_id` or `user` and immediately saw which policy was evaluated and why it was denied, without needing to page the identity team.

Versioned bundles also made rollbacks easier. When a new deployment policy accidentally blocked weekend rollouts, the engineering team didn't panic. They simply republished the previous bundle while debugging the change. For the first time, authorization had the same safety nets as any other critical system at ACME.

10.2.4 Continuous integration for policies

At ACME, validating with schemas, versioning, code reviews, and operational discipline are all integrated into their continuous integration (CI) pipeline. When Bob submits a pull request to the policy repository, it triggers a series of automated checks: syntax and schema validation, static analysis to find errors, and unit tests for expected behavior. Changes can only be merged if all checks pass.

From there, the pipeline created a signed bundle of Cedar policies and published it to the corresponding policy store. At runtime, the authorizer retrieved policies from that store to evaluate access requests. Figure 10.6 illustrates how ACME's CI/CD pipeline manages policy changes from authoring to runtime.